

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.

```
from google.colab import files
uploaded = files.upload() # This will prompt you to upload the file
```

Choose Files industrial_...l_dataset.csv

- **industrial_motor_rul_dataset.csv**(text/csv) - 64289 bytes, last modified: 4/30/2025 - 100% done

Saving industrial_motor_rul_dataset.csv to industrial_motor_rul_dataset.csv

```
import pandas as pd
import io
```

```
# Replace 'industrial_motor_rul_dataset.csv' with the actual uploaded filename
df = pd.read_csv(io.BytesIO(uploaded['industrial_motor_rul_dataset.csv']))
print(df.head()) # Display the first few rows
```

```
MotorSpeed Voltage Current Temperature NumberOfStarts RunTime \
0 1474.84 453.99 4.32 40.92 24 1403.99
1 1443.09 449.25 4.86 51.40 10 621.64
2 1482.38 440.60 4.21 55.86 8 1591.84
3 1526.15 433.53 4.69 78.88 21 1124.55
4 1438.29 446.98 3.11 65.57 6 1310.17

Vibration Age MotorLoad FaultHistory RUL
0 0.343 6343.29 54.22 0 76.17
1 0.343 2389.45 96.48 0 3925.04
2 0.860 4703.95 37.18 0 1307.49
3 0.672 6750.81 61.33 0 0.00
4 0.450 5751.76 78.88 0 367.80
```

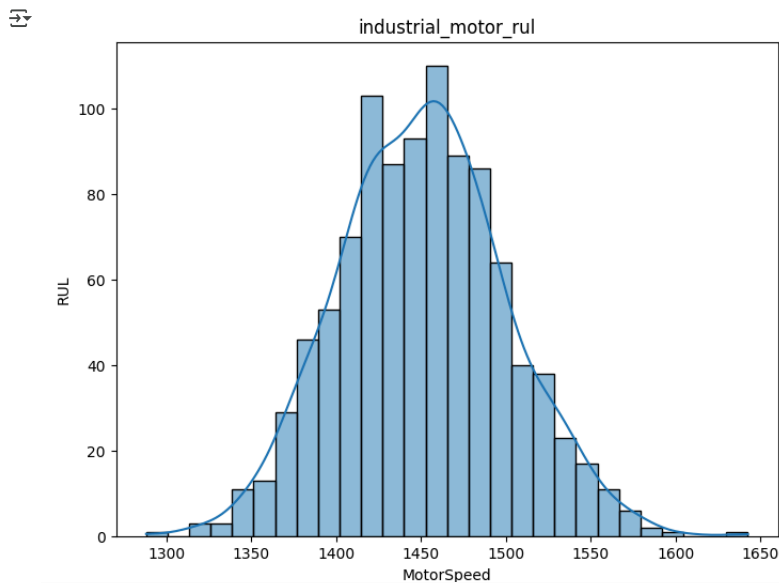
Generate

a slider using jupyter widgets

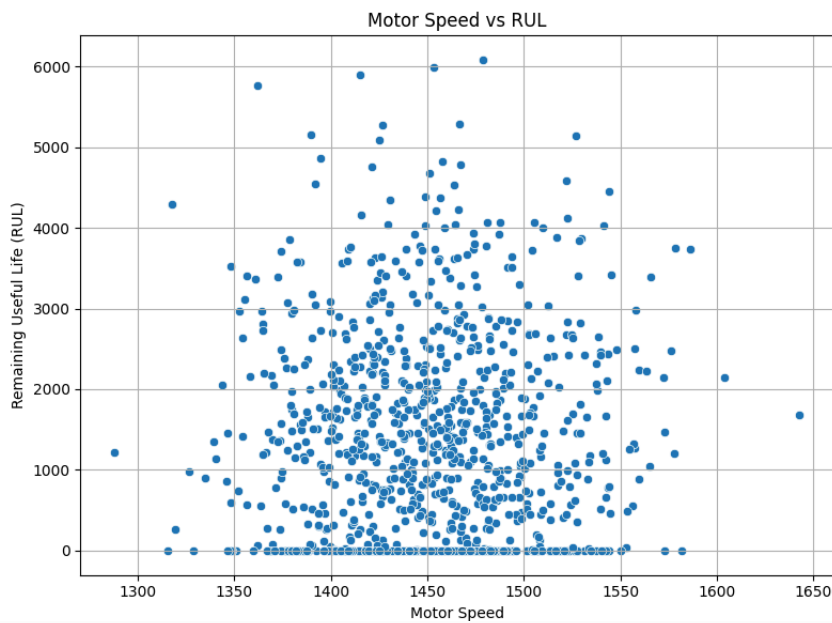


Close

```
import matplotlib.pyplot as plt
import seaborn as sns
# Chart 1: Histogram of Vibration
plt.figure(figsize=(8, 6))
sns.histplot(df['MotorSpeed'], kde=True) # KDE adds a smooth curve
plt.title('industrial_motor_rul')
plt.xlabel('MotorSpeed')
plt.ylabel('RUL')
plt.show()
```



```
plt.figure(figsize=(8, 6))
sns.scatterplot(data=df, x='MotorSpeed', y='RUL')
plt.title('Motor Speed vs RUL')
plt.xlabel('Motor Speed')
plt.ylabel('Remaining Useful Life (RUL)')
plt.grid(True)
plt.tight_layout()
plt.show()
```

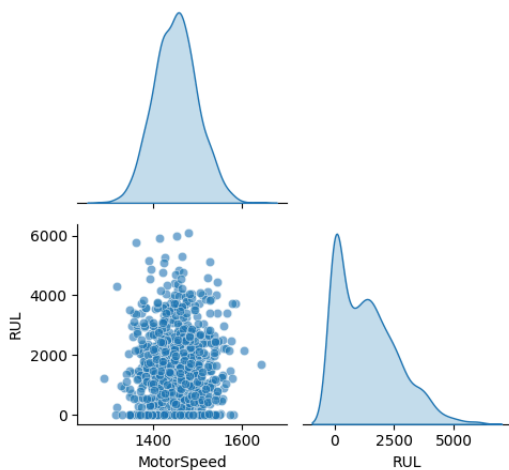


```
import seaborn as sns
import matplotlib.pyplot as plt

selected_columns = ['MotorSpeed', 'RUL']
sns.pairplot(df[selected_columns], corner=True, diag_kind='kde', plot_kws={'alpha': 0.6})
plt.suptitle('Pairwise Relationships Between Key Features', y=1.02)
plt.tight_layout()
plt.show()
```



Pairwise Relationships Between Key Features



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